The 27 Lines on a Smooth Cubic Surface

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Abstract

Every smooth cubic surface in $\mathbb{P}^3$ over an algebraically closed field contains exactly 27 lines. We give a short proof that runs through the incidence correspondence $Z \subset \mathbb{P}^{19} \times \mathbb{G}(1,3)$ between cubic surfaces and lines. First, we enumerate the 27 lines on the Fermat cubic explicitly. Second, we show that Z is a $\mathbb{P}^{15}$-bundle over the Grassmannian $\mathbb{G}(1,3)$, hence smooth of dimension 19. Third, a first-order deformation computation shows that a line on a smooth cubic surface has no infinitesimal deformations inside the surface. Together these facts make the projection $Z \to \mathbb{P}^{19}$ finite étale over the locus of smooth cubics, so the number of lines is the same for every smooth cubic surface, and the Fermat cubic shows that it equals 27. We also recover the number as the top Chern class $\int c_4(\mathrm{Sym}^3 S^\vee) = 27$ on the Grassmannian. We close with the lattice-theoretic picture (Pic and the E_6 root system) and with pointers to modern refinements: virtual fundamental classes, derived categories, and the $\mathbb{A}^1$-enumerative count of Kass and Wickelgren.

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1 Introduction

One of the most striking classical results in algebraic geometry is that every smooth cubic surface in $\mathbb{P}^3$ contains exactly 27 distinct lines. This theorem, due to Cayley and Salmon (1849), remains a fundamental example that illustrates key concepts in enumerative geometry, intersection theory, and moduli spaces. This note summarizes the classical derivation of this result and explores its broader implications in modern algebraic geometry.

The argument has three steps. In §2 we count the lines on the Fermat cubic by hand and set up the incidence correspondence Z, which we show is smooth of dimension 19. In §3 we show that a line on a smooth cubic surface admits no first-order deformations. In §4 we combine these facts into the theorem. The last three sections describe the structure of the configuration of lines and point to modern refinements; they are expository and quote results without proof.

Throughout this note, $k = \bar{k}$ is an algebraically closed field. For the explicit computation on the Fermat cubic we assume $\operatorname{char} k \neq 3$; Remark 4.2 gives an argument that works in every characteristic. Background on the Veronese embedding, Grassmannians, and Bertini's theorem can be found in [Vak; Har77].

2 The geometry of cubic surfaces

A cubic surface in projective space $\mathbb{P}^3$ is defined as the zero locus of a nonzero homogeneous polynomial of degree 3:

$$F(x_0, x_1, x_2, x_3) = 0. \tag{1}$$

Some important examples include:

- The Fermat cubic: $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$.
- The Clebsch diagonal cubic: $x_0^3 + x_1^3 + x_2^3 + x_3^3 = (x_0 + x_1 + x_2 + x_3)^3$.

If $\operatorname{char} k \neq 3$, the Fermat cubic is smooth, since its partial derivatives $3x_i^2$ vanish simultaneously only at the origin. As an example, we determine all lines on the Fermat cubic.

Proposition 2.1. *Assume* $\operatorname{char} k \neq 3$. *The lines on the Fermat cubic are exactly*

$$x_0 + \omega x_i = 0, \quad x_j + \omega' x_k = 0 \quad (\{i, j, k\} = \{1, 2, 3\},\ j < k), \tag{2}$$

where ω, ω' *are third roots of unity. Since there are three choices of* i *(each of which determines the pair* $j < k$*) and three choices each of* ω *and* ω'*, Eq. (2) produces* $3 \times 3 \times 3 = 27$ *distinct lines.*

Proof. Each line in (2) lies on the surface: on it, $x_0^3 = -\omega^3 x_i^3 = -x_i^3$ and likewise $x_j^3 = -x_k^3$.

Conversely, let $l \subset \mathbb{P}^3$ be a line on the Fermat cubic, that is, a 2-dimensional subspace $W \subset k^4$. For some pair of coordinates, the projection of W onto those two coordinates is an isomorphism. Permutations of the coordinates preserve the Fermat equation and permute

the lines (2), so we may assume that this pair is (x_2, x_3). Then l can be written as $x_0 = ax_2 + bx_3$, $x_1 = cx_2 + dx_3$. Plugging this into the equation $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$ and comparing the coefficients of x_2^3, x_3^3, $x_2^2x_3$, and $x_2x_3^2$ (the last two carry a factor 3, which is invertible because $\operatorname{char} k \neq 3$), one gets

$$a^3 + c^3 + 1 = b^3 + d^3 + 1 = a^2b + c^2d = ab^2 + cd^2 = 0. \tag{3}$$

For Eq. (3) to have a solution, at least one of a, b, c, d must vanish. Indeed, if all four were nonzero, dividing $a^2b = -c^2d$ by $ab^2 = -cd^2$ would give $a/b = c/d =: \lambda$, and then $a^2b + c^2d = \lambda^2(b^3 + d^3) = -\lambda^2 \neq 0$, a contradiction. If $a = 0$, then $c^3 = -1$, so $c \neq 0$, and $c^2d = 0$ forces $d = 0$ and hence $b^3 = -1$. The cases $b = 0$, $c = 0$, and $d = 0$ are analogous. Hence exactly two of the four numbers vanish, namely $\{a, d\}$ or $\{b, c\}$, and the other two are third roots of -1. Writing a third root of -1 as $-\omega$ with $\omega^3 = 1$, these are the lines $x_0 + \omega x_3 = x_1 + \omega' x_2 = 0$ and $x_0 + \omega x_2 = x_1 + \omega' x_3 = 0$, which are of the form (2). □

A conceptual way to think of general cubics is to consider the degree-3 Veronese embedding of $\mathbb{P}^3$,

$$\nu_3 : \mathbb{P}^3 \to \mathbb{P}^{19}, \qquad [x_0 : \cdots : x_3] \mapsto [\text{all 20 monomials of degree 3 in } x_0, \ldots, x_3]. \tag{4}$$

Then a cubic surface is identified with the intersection of $V = \nu_3(\mathbb{P}^3)$ with a hyperplane

$$H_c = \{\, [y_0 : \cdots : y_{19}] \,|\, c_0y_0 + \cdots + c_{19}y_{19} = 0 \,\}, \tag{5}$$

where $c_0, \ldots, c_{19}$ are the coefficients of F. The coefficient vector c is a point of the dual projective space; from now on we write $\mathbb{P}^{19}$ for this space of cubic forms and identify a point of it with the corresponding cubic surface. By Bertini's theorem, $V \cap H_c$ is a smooth codimension-1 subvariety of V for *generic* c. Here, 'generic' means that c is chosen from a dense open subset $U \subset \mathbb{P}^{19}$, the locus of smooth cubic surfaces.

Let $\mathbb{G}(1,3) = \mathrm{Gr}(2,4)$ be the Grassmannian of projective lines in $\mathbb{P}^3$, or equivalently, of planes in $\mathbb{A}^4$. We define the incidence correspondence

$$\begin{array}{ccc} & Z & \\ {\scriptstyle \mathrm{pr}_1}\swarrow & & \searrow{\scriptstyle \mathrm{pr}_2} \\ \mathbb{P}^{19} & & \mathbb{G}(1,3) \end{array} \qquad Z = \{\, (X, l) \,|\, X \supset l \,\} \subset \mathbb{P}^{19} \times \mathbb{G}(1,3) \tag{6}$$

It will be important that Z is a smooth scheme of dimension 19. Since $\dim Z = \dim \mathbb{P}^{19}$, one expects pr_1 to have finite fibers over a generic point, which means that a general cubic surface contains finitely many lines. Sections 3 and 4 establish this in a strong form. On the other side, $\mathrm{pr}_2 : Z \to \mathbb{G}(1,3)$ turns out to be a $\mathbb{P}^{15}$-bundle, which establishes that Z is smooth.

To see this, note that $\mathbb{G}(1,3)$ has a covering by copies of $\mathbb{A}^4$ given by row-reduced 2-by-4 matrices of full rank.

$$\begin{pmatrix} 1 & 0 & a & b \\ 0 & 1 & a' & b' \end{pmatrix} \rightsquigarrow l = \left\{ [s : t : sa + ta' : sb + tb'] \in \mathbb{P}^3 \,\middle|\, [s : t] \in \mathbb{P}^1 \right\} \in \mathbb{G}(1,3) \tag{7}$$

For instance, the collection of all lines l in (7) for varying a, a', b, b' forms a chart $\mathbb{A}^4$. Observe that up to a PGL_4 transformation, every line $l \in \mathbb{G}(1,3)$ can be mapped to the one in (7) with $a = a' = b = b' = 0$. Over this point, the fiber of pr_2 is the subscheme of $\mathbb{P}^{19}$ given by four equations

$$c_{\{000\}} = c_{\{001\}} = c_{\{011\}} = c_{\{111\}} = 0, \tag{8}$$

where we now write $c_0 y_0 + \cdots + c_{19} y_{19}$ as $c_{\{000\}} x_0^3 + \cdots + c_{\{333\}} x_3^3$, with indices running over multisets $\{ijk\}$. Eq. (8) confirms that the fiber of pr_2 over l is indeed $\mathbb{P}^{15}$.

The fibers are linear subspaces that vary algebraically with l. Over the chart (7), the condition $l \subset X$ says that the four coefficients of the binary cubic form $F(s, t, sa + ta', sb + tb')$ vanish. These are four linear forms in c whose coefficients are polynomials in (a, a', b, b'). By the PGL_4-homogeneity just used, they are linearly independent at every point of the chart. The kernel of a matrix of polynomials of constant rank is a vector bundle, so $\mathrm{pr}_2^{-1}(\mathbb{A}^4)$ is the projectivization of a rank-16 subbundle of the trivial bundle $k^{20} \times \mathbb{A}^4$. Hence pr_2 is a Zariski-locally trivial $\mathbb{P}^{15}$-bundle, and Z is smooth and irreducible of dimension $4 + 15 = 19$.

3 Deformation of lines

In this section, we turn to the analysis of pr_1 and show that over a smooth cubic surface X, the fiber $\mathrm{pr}_1^{-1}(X)$ has zero Zariski tangent space at each of its points. In particular, every point of the fiber is isolated and reduced.

Let l be a line in a smooth cubic surface X. After a PGL_4 transformation, we may again assume that l is the line (7) with $a = a' = b = b' = 0$, so that the equation F of X satisfies Eq. (8). We compare l with the nearby line l':

$$l = \left\{ [s : t : 0 : 0] \,\middle|\, [s : t] \in \mathbb{P}^1 \right\}, \tag{9a}$$

$$l' = \left\{ [s : t : sa + ta' : sb + tb'] \,\middle|\, [s : t] \in \mathbb{P}^1 \right\}. \tag{9b}$$

Proposition 3.1. *Let $R = k[\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3]/(\epsilon_i \epsilon_j : 0 \le i, j \le 3)$ with maximal ideal $\mathfrak{m} = (\epsilon_0, \ldots, \epsilon_3)$. Interpret a, a', b, b' as elements of $\mathfrak{m}$, so that l' is a first-order deformation of l. Then, for fixed smooth X, the line l' is contained in X only if $a = a' = b = b' = 0$. Equivalently, the Zariski tangent space of the fiber $\mathrm{pr}_1^{-1}(X) \subset \mathbb{G}(1,3)$ at l is zero.*

Proof. Assume that X contains l' for some $a, a', b, b' \in \mathfrak{m}$. We show that all four vanish. By Eq. (8), every monomial of F contains x_2 or x_3. Monomials of degree at least 2 in (x_2, x_3) vanish after the substitution, because the product of two elements of $\mathfrak{m}$ is zero. Hence, for $(x_0, x_1, x_2, x_3) = (s, t, sa + ta', sb + tb')$, we require that

$$\begin{aligned} F(x_0, x_1, x_2, x_3) &= x_2 \left(c_{\{002\}} x_0^2 + c_{\{012\}} x_0 x_1 + c_{\{112\}} x_1^2 \right) + x_3 \left(c_{\{003\}} x_0^2 + c_{\{013\}} x_0 x_1 + c_{\{113\}} x_1^2 \right) \\ &= (sa + ta') \underbrace{\left(c_{\{002\}} s^2 + c_{\{012\}} st + c_{\{112\}} t^2 \right)}_{=: C_2(s,t)} + (sb + tb') \underbrace{\left(c_{\{003\}} s^2 + c_{\{013\}} st + c_{\{113\}} t^2 \right)}_{=: C_3(s,t)} \end{aligned} \tag{10}$$

vanishes identically as a polynomial in s, t with coefficients in R. Furthermore, since X is smooth at every point of l, the gradient

$$\left(\frac{\partial F}{\partial x_0} \quad \frac{\partial F}{\partial x_1} \quad \frac{\partial F}{\partial x_2} \quad \frac{\partial F}{\partial x_3}\right)\Big|_{(s,t,0,0)} = \begin{pmatrix} 0 & 0 & C_2(s,t) & C_3(s,t) \end{pmatrix} \tag{11}$$

does not vanish at any $(s,t) \neq (0,0)$. In other words, the polynomials C_2, C_3 have no common zeros in $\mathbb{P}^1$. In particular, neither of them is identically zero: otherwise any zero of the other one, which exists because k is algebraically closed, would be a common zero. Since k is algebraically closed, each of C_2, C_3 therefore has either two distinct zeros or one double zero in $\mathbb{P}^1$.

We will use the following observation repeatedly. If $C_2(p,q) = 0$ at a point $[p:q] \in \mathbb{P}^1$, then $C_3(p,q) \in k^\times$, and evaluating (10) at $(s,t) = (p,q)$ gives $pb + qb' = 0$. Likewise, $C_3(p,q) = 0$ implies $pa + qa' = 0$. There are two cases.

i. Suppose one of C_2, C_3, say C_2, has two distinct zeros $[p_1:q_1] \neq [p_2:q_2]$. Then $p_1 b + q_1 b' = p_2 b + q_2 b' = 0$, and since (p_1, q_1) and (p_2, q_2) are linearly independent, $b = b' = 0$. Now (10) reads $(sa + ta')C_2(s,t) = 0$. Evaluating it at two distinct points $[p_3:q_3], [p_4:q_4]$ where C_2 does not vanish gives $p_3 a + q_3 a' = p_4 a + q_4 a' = 0$, so $a = a' = 0$.

ii. If both C_2, C_3 have double roots, these roots are distinct, because C_2 and C_3 have no common zero. Apply a PGL_2 transformation of the coordinates (s,t) on l, induced by a linear change of (x_0, x_1) that preserves the normal form above and transforms (a, a', b, b') linearly, so that $C_2(s,t) = \alpha s^2$ and $C_3(s,t) = \alpha' t^2$ with $\alpha, \alpha' \in k^\times$ (whence their roots are $[0:1]$ and $[1:0]$, respectively). Plugging the double roots into Eq. (10) yields $a = b' = 0$, and the remaining equation

$$\alpha a' s^2 t + \alpha' b s t^2 = 0 \qquad \forall s, t \tag{12}$$

implies $a' = b = 0$.

For the last claim, note that in the chart (7) the fiber $\mathrm{pr}_1^{-1}(X)$ is cut out by the coefficients of $F(s, t, sa + ta', sb + tb')$. A tangent vector to it at l is a solution with $(a, a', b, b') \in (k\epsilon_0)^4$, and such a solution is zero by what we have just shown. □

4 Counting the 27 lines

We now combine the three ingredients: Z is smooth of dimension 19 (§2), lines on smooth cubics do not deform (Proposition 3.1), and the Fermat cubic has 27 lines (Proposition 2.1).

Theorem 4.1. *Assume* $\operatorname{char} k \neq 3$. *Every smooth cubic surface* $X \subset \mathbb{P}^3$ *contains exactly 27 lines.*

Proof. Let $U \subset \mathbb{P}^{19}$ be the open subset of smooth cubic surfaces and $Z_U = \mathrm{pr}_1^{-1}(U)$.

pr_1 is étale on Z_U. Let $(X, l) \in Z_U$. The kernel of $d\,\mathrm{pr}_1 : T_{(X,l)}Z \to T_X \mathbb{P}^{19}$ is the Zariski tangent space of the scheme-theoretic fiber $\mathrm{pr}_1^{-1}(X)$ at l, which is zero by Proposition 3.1.

Hence $d\,\mathrm{pr}_1$ is an injective linear map between vector spaces of the same dimension 19, so it is an isomorphism. Since Z and $\mathbb{P}^{19}$ are smooth, pr_1 is étale at (X, l).

$pr_1|_{Z_U} : Z_U \to U$ is finite étale. It is proper, because Z is closed in $\mathbb{P}^{19} \times \mathbb{G}(1,3)$ and $\mathbb{G}(1,3)$ is projective. An étale morphism is quasi-finite, and a proper quasi-finite morphism is finite. Therefore $(\mathrm{pr}_1)_*\mathcal{O}_{Z_U}$ is a locally free sheaf on U. Its rank is constant because U, a nonempty open subset of $\mathbb{P}^{19}$, is irreducible and hence connected. Since every fiber over a closed point is reduced, this rank equals the number of lines on X for every $X \in U$.

The rank is 27. The Fermat cubic lies in U and contains exactly 27 lines by Proposition 2.1. □

Remark 4.2 (Schubert calculus)**.** *The same number can be computed without choosing an example. Let S be the tautological rank-2 subbundle on $\mathbb{G}(1,3)$. Restricting a cubic form to each line defines a section s_F of the rank-4 bundle $E = \mathrm{Sym}^3 S^\vee$ on the 4-dimensional Grassmannian, and its zero locus is $pr_1^{-1}(X)$. Proposition 3.1 says precisely that s_F vanishes transversally when X is smooth. Hence the number of lines is $\int_{\mathbb{G}(1,3)} c_4(E)$. If α, β are the Chern roots of $S^\vee$, then E has Chern roots 3α, $2\alpha+\beta$, $\alpha+2\beta$, 3β, and*

$$c_4(E) = 9\alpha\beta(2\alpha^2 + 5\alpha\beta + 2\beta^2) = 18\,c_1^2c_2 + 9\,c_2^2, \qquad c_1 = c_1(S^\vee) = \sigma_1,\ c_2 = c_2(S^\vee) = \sigma_{1,1}. \tag{13}$$

With $\sigma_1^2\sigma_{1,1} = \sigma_{1,1}^2 = 1$ this gives $18 + 9 = 27$ [EH16]. This route works in every characteristic, which removes the assumption $\mathrm{char}\, k \neq 3$ *from Theorem 4.1.*

5 The role of the Picard group

A key modern perspective comes from the Picard group $\mathrm{Pic}(X)$, which classifies divisor classes (see [Har77, Ch. V, §4] and [Dol12, Ch. 9]):

- A smooth cubic surface is isomorphic to the blow-up of $\mathbb{P}^2$ at six points in general position. Hence $\mathrm{Pic}(X) \cong \mathbb{Z}h \oplus \mathbb{Z}e_1 \oplus \cdots \oplus \mathbb{Z}e_6$, with $h^2 = 1$, $e_i^2 = -1$, and all other products zero, and $K_X = -3h + \sum_i e_i$.
- The intersection pairing on $\mathrm{Pic}(X)$ encodes how lines intersect on the surface. The lines are exactly the classes E with $E^2 = E \cdot K_X = -1$: the six exceptional curves e_i, the fifteen classes $h - e_i - e_j$, and the six classes $2h - \sum_{k\neq i} e_k$, for a total of $6 + 15 + 6 = 27$.
- The classes of the lines generate the rank-7 lattice $\mathrm{Pic}(X)$. The orthogonal complement $K_X^\perp \subset \mathrm{Pic}(X)$ is a rank-6 lattice which, with the sign of the form reversed, is isomorphic to the root lattice of E_6.
- The group of permutations of the 27 lines that preserve their intersection numbers is the Weyl group $W(E_6)$, of order 51840, connecting the configuration to Lie theory.

6 Toward derived and modern perspectives

While classical algebraic geometry studies these configurations in a geometric and enumerative way, modern perspectives introduce derived and homotopical techniques:

- Moduli of cubic surfaces: Smooth cubic surfaces up to projective equivalence form a $19-15=4$ dimensional family. Their deformations are unobstructed, since $H^2(X,T_X)=0$ and $h^1(X,T_X)=4$. Derived deformation theory becomes relevant for singular cubics and for moduli problems in which obstructions do appear.
- Derived intersection theory: For a singular cubic surface, the zero locus of s_F in Remark 4.2 can be non-reduced or even positive-dimensional; the cone over a smooth plane cubic contains a one-parameter family of lines. The derived zero locus still carries a virtual fundamental class of degree $\int c_4(E)=27$ [BF97].
- Derived categories: Since X is the blow-up of $\mathbb{P}^2$ at six points, $D^b(X)$ has a full exceptional collection of length $3+6=9$, obtained from Beilinson's collection on $\mathbb{P}^2$ and Orlov's blow-up formula.
- Homotopy theory: Over an arbitrary field k with $\operatorname{char} k \neq 2$, the count of lines can be refined to the $\mathbb{A}^1$-Euler number of E. Kass and Wickelgren show that the lines, counted with an arithmetic type in the Grothendieck–Witt group, sum to $15\langle 1\rangle+12\langle -1\rangle \in \mathrm{GW}(k)$ [KW21]. Its rank recovers the 27 lines over $\mathbb{C}$. Its signature recovers the real count, in which the number of hyperbolic lines minus the number of elliptic lines is 3.

7 Conclusion

The theorem on the 27 lines on a cubic surface is not just a classical curiosity; it serves as a gateway into deeper mathematical structures, from intersection theory to derived geometry. Understanding this classical example provides insight into modern topics such as derived categories, moduli spaces, and enumerative geometry in the presence of singularities.

Next steps. Readers interested in derived perspectives may explore Behrend–Fantechi's intrinsic normal cone and virtual fundamental classes [BF97], Illusie's cotangent complex [Ill71], or the role of Hochschild cohomology in moduli theory.

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